

Technology News

Technology to ease traffic woes

Traffic snarls make traveling by car quite frustrating. Take heart, there is a range of solutions on the horizon to help you avoid traffic jams.

Researchers at the University of California, Berkeley, have come up with an idea to gather information about vehicle speeds and traffic levels from drivers' GPS-enabled cellphones. The project, being run in partnership with phone company Nokia and digital map provider NAVTEQ, blends encrypted location data with traffic information from other sources before broadcasting the resulting travel information back to users' phones. Similar systems are already in commercial use in Europe.

In another feat, Microsoft Research has developed a system called JamBayes that continuously analyses current and historical traffic trends to make inferences about the future conditions of roads. It not only predicts future journey times and plots congestion-free routes but can also warn drivers to leave earlier in order to avoid impending gridlock.

Self-erasing documents

The colour-changing ability of nanoparticles could make it possible to create documents that wipe themselves clean after they've been read.

A team at Northwestern University in Evanston, Illinois, has coated gold nanoparticles with a layer of hair-like molecules called MUA. When zapped with ultraviolet light, these filaments change their shape and charge distribution, causing the nanoparticles to congregate together and change colour.

The colour of the nanoparticles depends on how close they are to one another. For instance, gold nanoparticles are red when separated, but become

violet, then blue, then colourless as they cluster together.

To put this colour-changing ability to good use, the team dispersed the nanoparticles in a gel and sandwiched it between plastic sheets to produce a thin, red film. When the researchers shone UV light at the film, they found they could print a range of images or write words onto the film in just a few seconds.

The colour change is not permanent, however. In the absence of UV light, the MUA gradually reverts to its original shape, allowing the nanoparticles to disperse and the images to disappear.

Artificial brain within ten years

A detailed, functional artificial human brain can be built within the next ten years, claims Henry Markram, director of the Blue Brain Project. Markram has already simulated elements of a rat brain. According to him, a synthetic human brain would be of particular use in finding treatments for mental illnesses.

The Blue Brain project at Switzerland's EPFL (École Polytechnique Fédérale de Lausanne) aims to reverse engineer the mammalian brain from laboratory data. In particular, the research team has focused on the neocortical column—repetitive units of the mammalian brain known as the neocortex. The project now has a software model of tens of thousands of neurons—each of which is different—which has allowed them to digitally construct an artificial neocortical column.

Although each neuron is unique, the team has found that the patterns of circuitry in different brains have com-

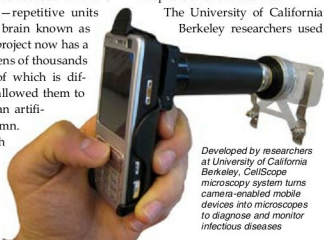
mon patterns. To make the model come alive, the team feeds the models and a few algorithms into a supercomputer. Simulations have started to give the researchers clues about how the brain works. For example, they can show the brain a picture and follow the electrical activity in the machine.

Phone add-on to diagnose TB

Researchers have developed an add-on to a mobile phone that can take detailed images and analyse them to diagnose diseases such as tuberculosis (TB). The gadget, called CellScope, is made up of conventional microscope optics as well as some equipment to make it function as a fluorescence microscope.

Fluorescence occurs when certain molecules are illuminated with a certain colour and 'shine' for a period in a different colour. Diagnosing tuberculosis requires a fluorescence microscope, which can illuminate a blood sample that has been treated with fluorescent 'tagging' molecules and detect just the light that those molecules emit with great sensitivity. (Fluorescent 'tagging' molecules are specially designed to latch on to, for instance, the bacteria that are a sign of tuberculosis.) However, typical fluorescence microscopes are bulky, expensive devices limited to hospitals and laboratories.

The University of California Berkeley researchers used



Developed by researchers at University of California Berkeley, CellScope microscopy system turns camera-enabled mobile devices into microscopes to diagnose and monitor infectious diseases